MINING DEALER BEST PRACTICE

C175 Cylinder Block Flushing

Maintenance and Repair | Application | Component Life Management | **Component Renewal (CRC)** | MARC Management

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1.0 Introduction

This Best Practice is for all Global CRC's, that are looking to further improve and certify C175 cylinder block cleanliness, prior to assembly.

This tooling was locally design and fabricated by Matco, to provide closed loop flushing, of all the C175 cylinder block oil ports, at regulated pressures, of up to 50 PSI.

The mobile cart design is equipped with multiply drawer storage, 60 Liter fluid tank, 50 GPM diaphragm pump, with three individual valves. Two valves will either direct fluid to the upper or lower sections, of the cylinder block or a combined supply and return to the tank. The third valve is utilized for kidney looping of the tank fluid, to maintain a fluid cleanliness of 16/13 or less.

The supply fluid from the pump to the cylinder block, goes through a two-micron fuel filter and the returning fluid from the cylinder block goes through the (patch sample) and then through the return two-micron fuel filter to the tank.

Figure 1. C175-20 Flushing Tooling

2.0 Best Practice Description

Having closed loop flushing capabilities, to constantly supply clean fluid flow and pressure through all oil ports of the C175 cylinder block, and monitoring (patch samples) of the return fluid during the flushing process, which will ensure that all cylinder block oil ports are completely clean, and the cylinder block is certified for assembly.

3.0 Implementation Steps

3.1 Defining the Scope: With the volume of C175-20 engines Matco rebuild annually, it was decided to start with the C175, due to the financial impact, if the Customer expected engine life was not achieved, or if any premature failure occurred, from hidden debris and Matco will continue to develop this tooling for the other engine models they rebuild.

3.2 Analysis for Design: Once the scope was defined by the Matco design team, it was then conceptually laid out in drawings, identifying, mobile cart size, tool storage needs, fluid tank and pump capacity, valving, filtration, tube lengths and interface connections.

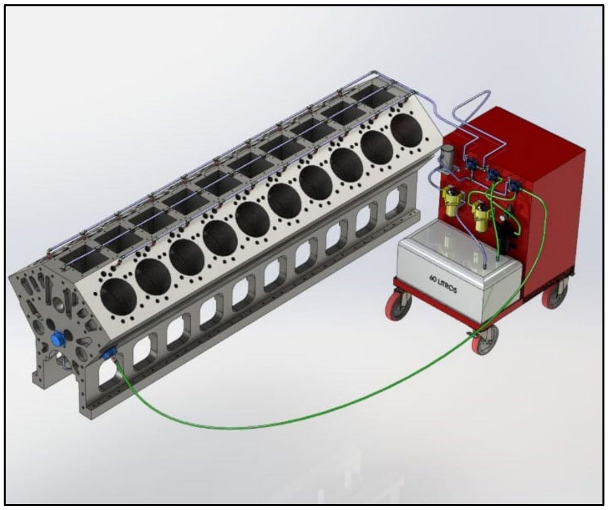

Figure 2. C175 Flushing Tool Group Design.

3.3 Flushing Tooling Group: This photo briefly illustrates some of the main features of the flushing cart and layout of the fluid tank, valving, inline filters, sample patch cylinder, diaphragm pump, line routing, etc.

Figure 3. C175 Flushing Loop Tool Group.

3.4 Connections that were not available on the local market (i.e.) for the spray-jet ports, main bores, camshaft cross over ports, were design and then fabricated in Nylamid material (Illustration below) utilizing O-rings grooves for secure sealing and the camshaft port, cross over connection was fabricated in silicon (blue) with an Internal tube passage and the mold was design by the team and fabricated in the 3D printer (photos of fabricated samples below).

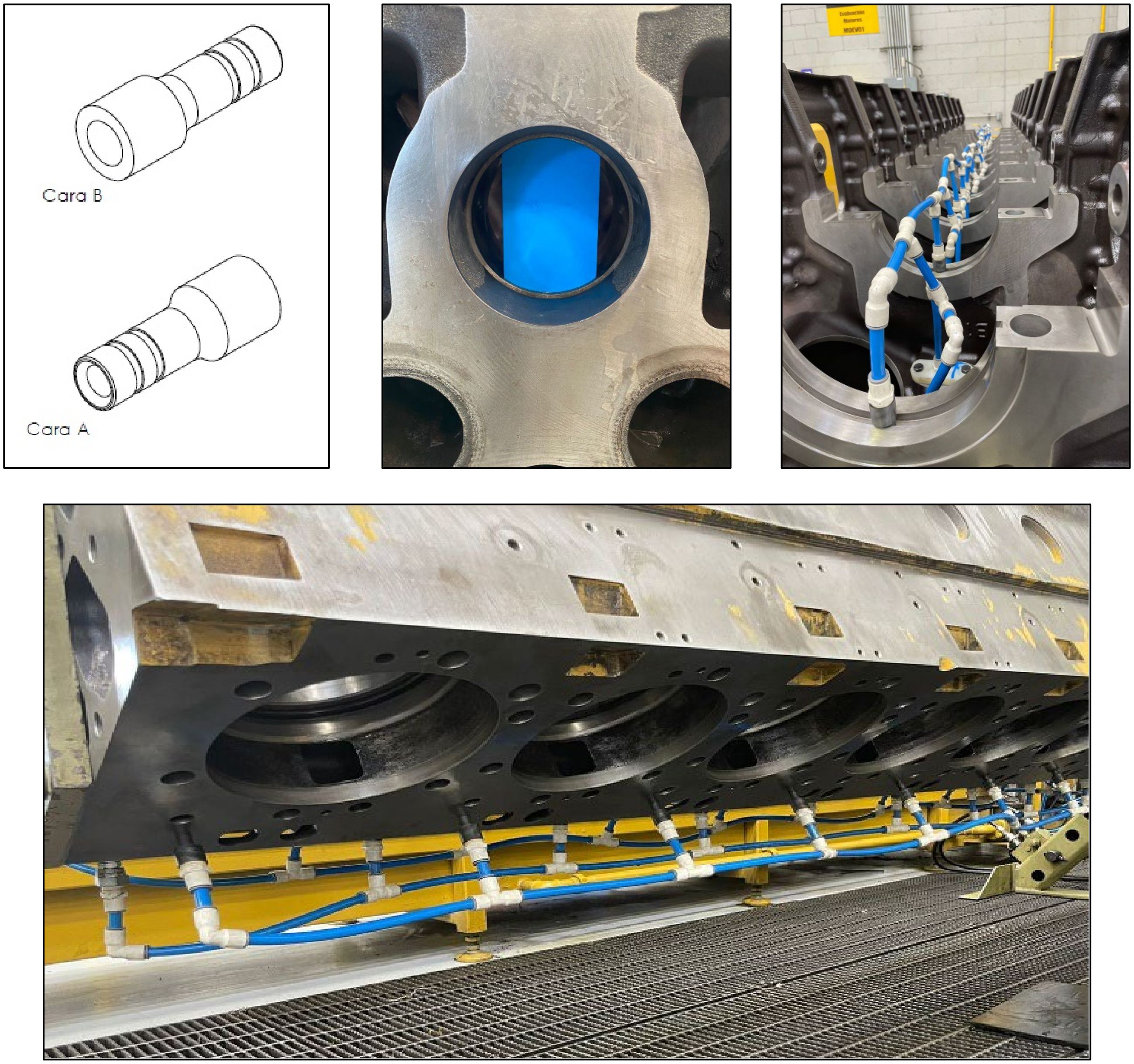

Figure 4. Nylamid connectors for oil ports, camshaft / valve train ports.

3.5 The majority of the customized items, that could not be locally purchased were fabricated by the Matco team, with minor outsource support and the man-hours, to fit up this tooling on a C175-20 engine is approximately 1.5 hours.

3.6 Fit-Up Process.

- Option to install cylinder block in rollover fixture.
- Install all connectors to (i.e.) main bearing bore, cooling jet, camshaft, valve train ports, etc. in the cylinder block.
- Connect the supply and return hoses from the tooling to the cylinder block.
- Install the patch.
- Start the pump and set it to approximately (25-35 PSI) checking for any leaks and if no leaks are visible, the system pressure can be increased to a maximum of (45-50 PSI).
- After approximately five minutes of flushing, turn off the pump and check the patch for any debris.
- If no debris, is identified in the patch. Remove the tooling.
- If any debris is identified in the patch. Repeat the process until we can observe a thoroughly clean patch.

4.0 Benefits

- Pressure flush of all cylinder block oil ports.
- Patch sample taken and recorded in history file, of oil port cleanliness before assembly.
- Critical oil ports in the cylinder block are thoroughly clean, ensuring no bearing damage from debris at start-up.

5.0 Resources Required

- Flushing Loop Tooling.
- Hoses, connectors, fittings, plugs, etc.
- Engine Stand

6.0 Supporting Attachments / References

- M0080689 / Cylinder Block Cleaning and Audit Process

Please contact the Matco design team for more details as there is an update to the prototype in progress as of the submission date of this Best Practice.

7.0 Related Best Practices

N/A

8.0 Acknowledgements

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